

Ayoung Park

Financial Management Stabilization Department, Financial Supervisory Service
Seoul 07321, Republic of Korea
+82-10-2100-1617
ayoungp@fss.ok.kr, ayppark@gmail.com
<https://ayppark.github.io>

WORK EXPERIENCE

Senior Manager Jan 2026 - present
Supervisory Strategy Research Team, Supervision Innovation Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on blockchain-based financial networks, regulatory frameworks, and supervisory innovation (including AML and crypto-assets).

Senior Manager Mar 2024 - Jan 2026
Future Finance Research Team, Financial Market Stabilization Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on Financial Market Analysis related with the financial stability, regulation, and innovation.

Research Assistant Professor July 2021 - Feb 2024
The Institute of Knowledge Management, Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Conducted research on Financial Market Analysis via Connectedness and Deep Learning with the 5-years funding from National Research Foundation(NRF) of Korea

Visiting Scholar Jan 2023 - Feb 2023
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the prediction of the speed of collapse and recovery in the 2007-2009 global financial crisis on international stock markets via complex network theory.

Lecturer Mar 2021 - June 2021
Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Played a pivotal role in the conceptualization and creation of the *Introduction to Fintech* course, ensuring its relevance and applicability in the current financial landscape.

Visiting Scholar Mar 2019 - Feb 2020
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the general mechanism of response and recovery between anesthesia on the brain and the impact of the 2007-2009 global financial crisis on international stock markets.

Visiting Scholar Jan 2019 - Feb 2019
Department of Physics, Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea

- Research on the non-linear dynamics and complexity based on brain and financial market.

Visiting Scholar Jan 2017 - Feb 2017
 Division of Business Administration, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, United States

- Research on the role of board diversity and behavior finance.

Education

Ph.D. in Finance Feb 2021
 Chosun University, Gwangju, Republic of Korea

- Dissertation title: Essays on Financial Connectedness, Systemic Risk, and Performance.
- Excellent Research Award

M.S. in Finance Aug 2016
 Chosun University, Gwangju, Republic of Korea

- Thesis title: The Empirical Study on Measuring Systemic Risk using Variance Decomposition and Debrank method

B.Sc. in Mathematics and B.A. in Business Administration Feb 2015
 Chosun University, Gwangju, Republic of Korea

- Summa Cum Laude

PROFESSIONAL TRAINING Anti-Money Laundering(AML) Specialist Program, Hanyang University, Dec 2025 – Focus on AML regulation, crypto-assets, and financial surveillance

RESEARCH INTERESTS Computational Finance, Risk Management, Financial Intermediation, Empirical Asset Pricing, Corporate Governance, Complex System, Graph Neural Network, Natural Language Processing

RESEARCH FUNDING

Title: *Financial Market Analysis via Connectedness and Deep Learning.*

- **Objective:** Investigating the connectedness among financial institutions and boards of directors from international countries. The focus lies on risk propagation, knowledge spillover, and its predictive potential for technological innovation.
- **Source:** National Research Foundation of Korea (NRF).
- **Amount:** A grant of 200 million KRW.
- **Duration:** July 2021 - Mar 2024.

WORKING PAPERS

- **Global Banking Networks and Loan Pricing**, Job Market Paper
 - **Best Paper Award, Semi-finalist in Financial Intermediation & Markets at Financial Management Association (FMA) 2023**
 - **Best Student Paper Award** at Fall conference of Korea Academy of Complexity Studies 2018 with a prize of 200,000 KRW (approx. 181 USD)
 - **Computing in Economics and Finance (CEF) 2023**
 - **Workshop of Asia- Pacific Applied Economics Association, 2023**
- **Stablecoin Prices as Shadow FX Quotes: Dollar News and the Korean Crypto Premium**
 - **Financial Management Association Annual Meeting, 2026**
- **Evaluating Model Warnings under Limited Review Capacity**

- **Investor Sentiment and Stock Return Predictability: The Influence of Online Stock Discussion Boards**, (with Danbi Park, Gabjin Oh)

JOURNAL ARTICLES

See also [my google scholar](#) page.

- Uncheol Lee, Hyoungkyu Kim, Minkyung Kim, Gabjin Oh, Pangyu Joo, Ayoun Park, Jamie Sleigh, Katie Warnaby, Iren Tracey, George Mashour, [Proximity to Explosive Synchronization Determines Network Collapse and Recovery Trajectories in Neural and Economic Crises](#), *PNAS*, (SCI, impact factor: 9.1, Ranked 14/135 in Multidisciplinary Sciences), Sep 2025.
- Ayoun Park and Gabjin Oh, [The Contagion Effect of Heterogeneous Investor Groups](#), *Plos One* (SCI, impact factor: 3.752, Ranked 29/134 in Multidisciplinary Sciences), Oct 2023.
- Ayoun Park and Gabjin Oh, [The Core of the Board Networks and Firm Value](#), *Frontiers in Physics* (SCI, impact factor: 3.718, Ranked 42/111 in Physics, Multidisciplinary), Feb 2023.
- Ayoun Park and Gabjin Oh, [Economic System Research by Using Statistical Physics](#), *Physics & High technology* (Highlighted Article), Aug 2022.
- Jinjoo Yoon, Ayoun Park, and Gabjin Oh, [Grouping Analysis of Social Media in Korean Financial Market](#), *New Physics: Sae Mulli* (SCOPUS), Feb 2022.
- Ayoun Park and Gabjin Oh, [Lending Diversification and Interconnectedness of Syndicated Loan Market](#), *Frontiers in Physics* (SCI, impact factor: 3.56, Ranked 30/110 in Physics, Multidisciplinary), Jan 2021.
- Jie Deng, A-young Park, and Gabjin Oh, [The Relationship between Cash Flows and Investment according to Economic Status](#), *Journal of the Korean Physical Society* (SCI, impact factor: 0.649, Ranked 87/111 in Physics, Multidisciplinary), Jan 2020.
- Ayoun Park and Gabjin Oh, [Systemic risk and Expected return](#), *New Physics: Sae Mulli* (SCOPUS), June. 2018.
- Ayoun Park and Gabjin Oh, [An Analysis of the Relationship between Market Risk and Information Flows among Trading Volume of the Industry Sector in the Korean Stock Market](#), *New Physics: Sae Mulli* (SCOPUS), May. 2017.
- Gabjin Oh, Ho-Yong Kim, and Ayoun Park, [Analysis of technological innovation based on citation information](#), *Quality and Quantity* (SSCI, impact factor: 1.25, Ranked 42/96 in Social Sciences, Interdisciplinary), Dec. 2016.
- A Young Park, Ho-Yong Kim, and Gabjin OH, [An Empirical Study on Measuring Systemic Risk Based on Information Flows using Variance Decomposition and DebtRank](#), *The Korean Operations Research and Management Science Society*, (KCI), Nov. 2015.

WORK IN PROGRESS

- Forecast Compression and Borrower-Size Gradients in Syndicated Lending
- Holdings Embeddings and Subsequent Illiquidity
- Missing Carbon and Model Risk
- Predictable, Not Deployable: Economic Networks and the Transportability of Housing Forecasts

**TEACHING
EXPERIENCE**

Introduction to Fintech, undergraduate, Chosun University (Spring 2021-2023)
Security Investment Practice, undergraduate, Chosun University (Spring 2023)
Management Statistics, undergraduate, Chosun University (Fall 2021, 2023)
Management and Economics, undergraduate, Chosun University (Fall 2021-2023)

**SPECIAL
ACHIEVEMENTS**

Awards

- *Best Paper Award*, Semi-finalist in Financial Intermediation at Financial Management Association (FMA), 2023
- *Ph.D. Excellent Research Award*, Chosun University, 2021
- *Best Research Award* for "Measuring Information Asymmetry of Inter-bank Network from Syndicated Loan Market", Fall Conference of Korea Academy of Complexity Studies, 2018
- *Best Research Award* for "Diversity in the Board Network Impact on Firm Performance", Fall Conference of Korea Academy of Complexity Studies, 2017
- *Encouragement Award* for "Analysis of Financial Stability Based on Investor Trading Behavior", 4th Conference of Social Sciences Korea Networking, 2017
- *Fellowship*, Brain Korea 21+, Korean Government, 2015-2020
- *Summa Cum Laude*, Department of Mathematics, Chosun University, 2014
- *Scholarship*, Catholic Archdiocese of Gwangju, 2014
- *Scholarship*, KT&G Scholarship Foundation, 2013-2014
- *Full-Tuition Scholarship*, Korea Student Aid Foundation (KOSAF), 2011-2014

Industry Experience

- *Researchers*, Technological Innovation and Data Construction for Financial/Economic Digital Twins in Companies and Industries at ETRI(Electronics and Telecommunications Research Institute), 2022- 2023

Volunteer Experience

- *Global Volunteer*, Happy Move funded Hyundai and Habitat: Participated in community house-building in the Philippines and intercultural exchange, Feb 2014
- *KT&G 7st SangSang Univ*, University Student Planning Group, 2013
- *Education*, Elementary school in Catholic, 2011-2014.

Invited Lectures

- Global Banking Network and Syndicated Loan Prices, at workshop of Asia-Pacific Applied Economics Association, Xian Jiaotong University, July. 2023
- Network Analysis in the Financial System, Invited Talk at conference of Korea Academy of Complexity Studies, Sookmyung Women's University, Dec. 2022; at the Ajou University, Oct. 2022; at the Pukyong National University, Aug. 2022
- From Research to Regulatory Affairs, Invited Talk at Inha University, Nov. 2025

PROFESSIONAL INTERVIEWS

- Invited Interview in [National Research Foundation of Korea\(NRF\)](#), Oct 2021 (based on the high score of research project)

CONFERENCE CONTRIBUTIONS

- Global Banking Network and Syndicated Loan Prices, *Sydney Banking and Finance Conference (SBFC) 2025*, *2023 Financial Management Association (FMA)*, *29th International Conference Computing in Economics and Finance (CEF)*, *2023 Fall Conference of Korean Finance Association*, *15th Asia Pacific Physics Conference*, *24th Interdisciplinary KASBA conference*, *Networks 2021: A Joint Sunbelt and NetSci Conference*, *2019 The Network Science Society*
- A Network Approach to Optimal Portfolio: Application to the Korean Stock Market, *29th International Conference Computing in Economics and Finance (CEF)*
- Measuring Similarity between trend Behaviors of Multivariate Time Series, *2023 European Financial Management Association (EFMA)*, *28th International Conference Computing in Economics and Finance (CEF)*
- Evolution of Social media activity-financial market Based on the complex network method, *28th International Conference on Statistical Physics*
- The Effect of Knowledge Diversity of Board Networks on Corporate Value, *27th International Conference Computing in Economics and Finance (CEF)*, *2018 Econophysics Colloquium*, *2018 Asia Pacific Econophysics Conference*, *2017 Fall conference of Korea Academy of Complexity Studies*
- Analysis of Mutual Fund Similarity Network in Korea Equity Market, *2018 Asia Pacific Econophysics Conference*, *The 13th INFINITI Conference on International Finance*
- Financial Stability Based on Information Flows among Investor Trading Behavior, *25th International Conference Computing in Economics and Finance (CEF)*, *2018 Asia Pacific Econophysics Conference*, *NetworkX-2018: International School and Conference on Network Science*, *2017 International Conference on SigmaPhi*, *2017 4th conference of Social Sciences Korea Networking (SSK)*, *2016 Japanese Association of Financial Econometrics and Engineering (JAFEE)*
 - Encouragement student paper award at the 4th conference of Social Sciences Korea Networking (SSK).
- Usefulness of the Fama and French three-factor model in Shanghai A-share market, *2016 International School and Conference on Network Science*, *2016 Fall conference of Korea Academy of Complexity Studies*

PROFESSIONAL SERVICE

- **Journal Referee:** Plos One, Journal of the Korea Society Industrial Information System, Humanities and Social Sciences Communications
- **Conference Discussant:** Fall conference of Korean Finance Association 2022; European Financial Management Association (EFMA) 2023; Financial Management Association (FMA) 2023, Asia Financial Management Association (Asia FMA) 2024
- **Conference Reviewer:** Financial Management Association (FMA) 2023
- **Conference Session Chair:** Computing in Economics and Finance (CEF) 2022, 2023; Financial Management Association (FMA) 2023; Asia Financial Management Association (Asia FMA) 2024

- **Professional Membership:** Financial Management Association (FMA), Korean Security Association, Korea-America Finance Association(KAFA)

SKILLS

Python, Stata, Matlab, C++, Latex, Gephi

LANGUAGES

English (Fluent), Korean (Native)

REFERENCE

- **Gabjin Oh, Ph.D.**, Professor, Division of Business Administration, Chosun University, Gwangju, Republic of Korea, *phecogjoh@chosun.ac.kr*
- **Uncheol Lee, Ph.D.**, Associate Professor, Department of Anesthesiology, Associate Director, Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI, United States, *uclee@med.umich.edu*
- **Woo-Sung Jung, Ph.D.**, Professor, Department of Industrial and Management Engineering/Physics, Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea, *wsjung@postech.ac.kr*
- **O-Kyu Kwon, Ph.D.**, Senior Researcher, NIMS-National Institute for Mathematical Sciences, South Korea, *okyu.kwon@gmail.com*
- **Hyeng Keun Koo, Ph.D.**, Professor, Department of Financial Engineering, Ajou University, Suwon, Republic of Korea, *hkoo@ajou.ac.kr*